Harry Nguyen

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Education

Georgia State University **Atlanta, GA**

Bachelor of Science in Computer Science, GPA: 3.75/4.0 *Expected May 2027*

- **Coursework:** Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures

Work Experience

Google | Software Engineer Intern – Chrome Browser May 2025 – Aug 2025

- Shipped a **C++ IPC transport layer** to Chrome's stable channel – serving **3B+ active users** at **sub-50ms p99** and **10K+ req/sec** – by choosing Protocol Buffers over custom serialization for schema evolution and cross-language compatibility
- Turned a visibly sluggish settings search into an instant one for power users – **96% p99 latency cut** (1,200ms down to sub-50ms), zero production regressions – by replacing a mutex-contended search path with a **lock-free concurrent trie**
- Verified linearizability of the lock-free trie under concurrent operations before shipping to Chrome stable, alongside design docs and code reviews with the Chrome infrastructure team
- Ended cross-team cascading UI bugs and accelerated feature delivery **68%** across **25K+ lines** of Chromium, at **95% test coverage**, by architecting an event-driven TypeScript/React system
- Landed production-ready changes into Chrome's codebase – reviewed and adopted by senior engineers – by delivering design docs and code reviews at **95% test coverage** alongside the Chrome infrastructure team

TiMoto AI | Software Engineer Sep 2025 – Present

- Cut incident root-cause time from unknown to **under 15 minutes** by building TiMoto's observability stack from zero – deployed Prometheus and Grafana metrics alongside CloudWatch dashboards across the gRPC and ML serving layers
- Restored **100% evaluation success rate** at **sub-50ms p99** by resolving a production **gRPC deadlock** – traced a shared resource acquisition conflict under concurrent calls and replaced inconsistent lock ordering with a single enforced sequencing rule
- Eliminated **OOM failures** under concurrent production load by replacing naive batching with a vLLM/PagedAttention inference engine, solving KV cache memory fragmentation
- Prevented single-zone failures from taking production down, sustaining **99.9% uptime** and cutting infrastructure cost **44% ($40-60/month)**, by architecting a multi-AZ ECS Fargate deployment with Terraform IaC and an auto-rollback circuit breaker pattern
- Cut mean-time-to-identify **68%** by building a post-mortem practice with runbooks and a shared on-call rotation

Develop for Good | Software Engineer Intern May 2024 – Aug 2024

- Cut API response time from 3+ seconds to **sub-100ms** on 10,000+ record datasets by diagnosing an N+1 query bottleneck and redesigning the data layer with **PostgreSQL indexing**
- Scaled a backend to support **500+ concurrent users** and cut deployment time 90% by designing a stateless AWS BaaS with JWT auth and a GitHub Actions CI/CD auto-scaling policy

Personal Projects

Pulumi – Contributor | ***Go, TypeScript, Infrastructure-as-Code (IaC)***

- Analyzed **Raft/Paxos consensus** in a distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures
- Submitted Go CLI features and bug fixes enabling **multi-cloud (AWS/Azure/GCP)** provisioning – under active review by core maintainers

Technical Skills

Systems & Concurrency: C++, lock-free data structures, linearizability, IPC, Protocol Buffers, UNIX/Linux, gRPC, exactly-once semantics
Languages: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust
Distributed Systems: gRPC, Circuit breakers, Kafka, Multi-AZ, Event-driven architecture, Raft/Paxos
ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, PyTorch, LangChain
Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP
AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor